$$\hat{H} = \hat{H}_{\mathcal{S}} + \hat{H}_{\rm int} = -\frac{1}{2}\Delta_0\hat{\sigma}_x + \frac{1}{2}\hat{\sigma}_z \otimes \sum_{i=1}^{N} g_i \hat{\sigma}_z^{(i)} \equiv -\frac{1}{2}\Delta_0\hat{\sigma}_x + \frac{1}{2}\hat{\sigma}_z \otimes \hat{E}. \quad (5.150)$$